Figure 1:Raw image

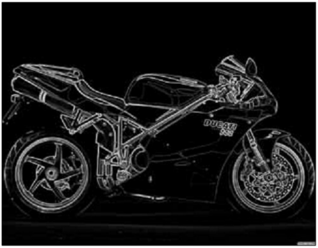

Figure 2: First image

-1	0	1
-1	0	1
-1	0	1

Vertical matrix

1	1	1
0	0	0
-1	-1	-1

Sobel: This is similar to Prewitt, but the differentiation matrices are different.

Canny: Canny edge detection is an optimal edge-detection algorithm. It uses multiple steps to achieve high precision.

Canny edge detection

Canny edge detection is a four-step process.

1. Initially, a Gaussian blur is applied to clear any speckles and free the image of noise. This step is optional. However, if the image has too much noise, or if you want

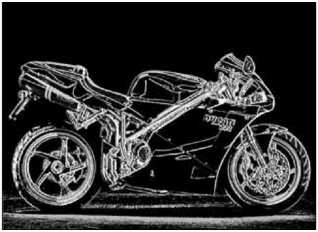

Figure 3: Angle map

to detect fewer edges, a Gaussian blur is mandatory.

2. A Sobel or Prewitt operator is applied to get two edge maps—the horizontal edge map *gradx* and the vertical edge map *grady.* The edge map is a combination of these two edge maps. Also, the direction of the gradient at each point, i.e., the angle, is calculated as follows:

```
theta = arctan ( grady / gradx )
```

To detect only the major edges, all edges below a certain threshold will be removed. For example, all pixels whose value is less than 10 will be made zero, indicating that it is not a pixel on any edge.

3. A non-maximum suppression algorithm is now applied. For a given pixel, there are eight neighbouring pixels. Hence, there are four possible directions: 0 degrees, 45 degrees, 90 degrees and 135 degrees. The angles are quantised into these four angles. Now consider a pixel in the image. Let us assume that the angle of gradient at that point is 0 degrees. This implies that the direction of the edge should be perpendicular to it, i.e., running along the 90 degrees line. Hence, let's check for the left and right neighbouring gradient values. If the gradient at the present pixel is greater than its left and right neighbours, then it is considered an edge; else, it is discarded. This process gives single-pixel-thick edges.
4. Now a hysteresis-based edge scanning is done. It is known that major important edges will be along curves, not as isolated points. A two-threshold scanning is done. The high threshold, *Th,* will decide the starting of the edge, and the *Tl* will decide the ending of the edge. Once this process is done, you get a binary map of the edges.

Implementation in Python

Let us see how this is implemented in Python using SciPy.